

Experiment 2

Creation and Network Configuration of Virtual Machines

VirtualBox + Ubuntu VMs + NAT Network

Student Name	Babin Bid
Roll Number	UG/02/BTCSE/2023/114
Registration Number	AU/2023/0009748
Section & Stream	Section B B.Tech CSE (Core)
Course & Semester	Cloud Computing Lab (CSE12045) · 7th Semester
Course Instructor	Prof. Anusuya Bera (Dept. of CSE)
Institution	ADAMAS University, Kolkata

1. Objective

Create a second Ubuntu virtual machine by cloning the existing Ubuntu server VM, configure both VMs to use the same VirtualBox NAT Network, verify their assigned IP addresses, and test communication between the two virtual machines using the ping command.

2. Requirements

Oracle VirtualBox, an existing Ubuntu server VM, an Ubuntu client VM/clone, and a configured VirtualBox NAT Network.

3. Procedure

Step 1 — Create the Ubuntu Client Virtual Machine

Open Oracle VirtualBox and create a new Ubuntu client VM. Select the Ubuntu ISO image and enable the unattended installation option as shown in the captured setup.

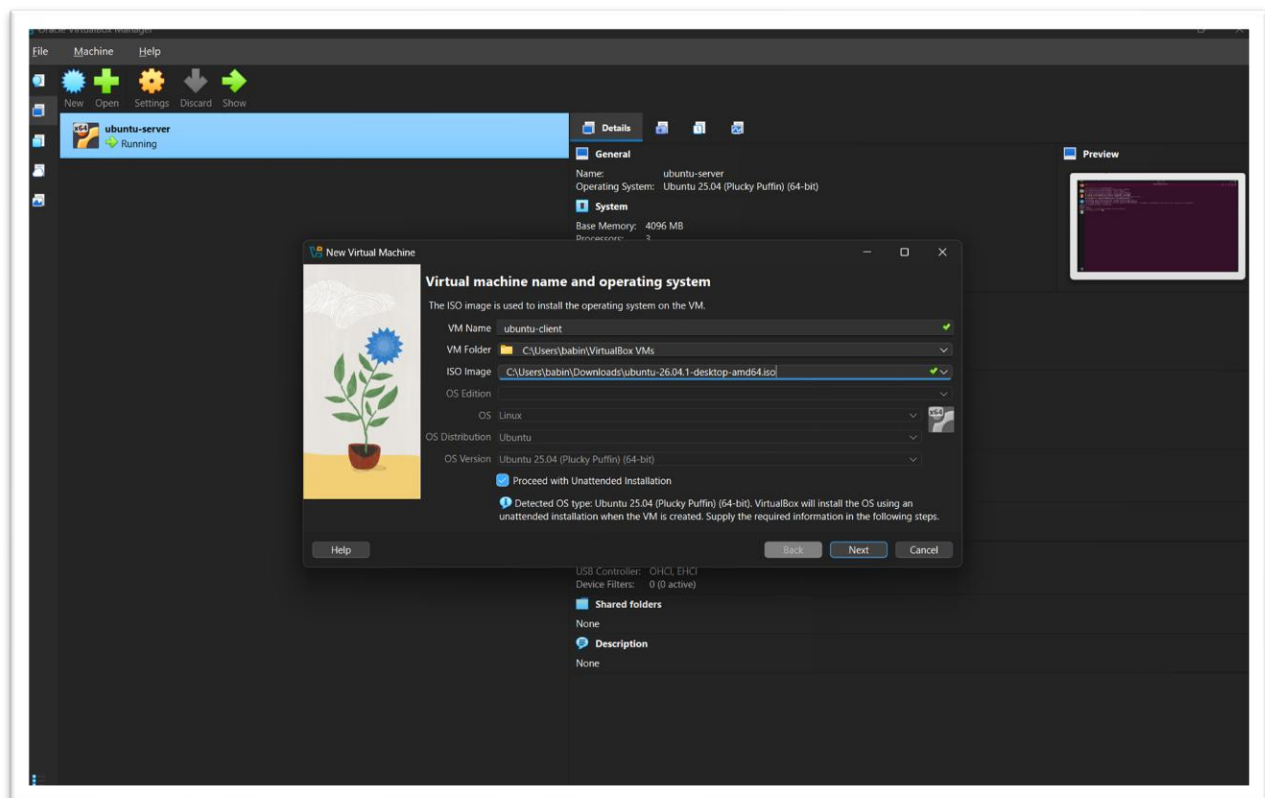


Figure 1 — VirtualBox New Virtual Machine window for the Ubuntu client.

Step 2 — Clone the Existing Ubuntu Server

An existing Ubuntu server VM can also be duplicated using the Clone Virtual Machine option. Select Full Clone so that an independent copy of the virtual machine is created. Use a different VM name, such as ubuntu-client (clone), and generate new MAC addresses for the network adapters.

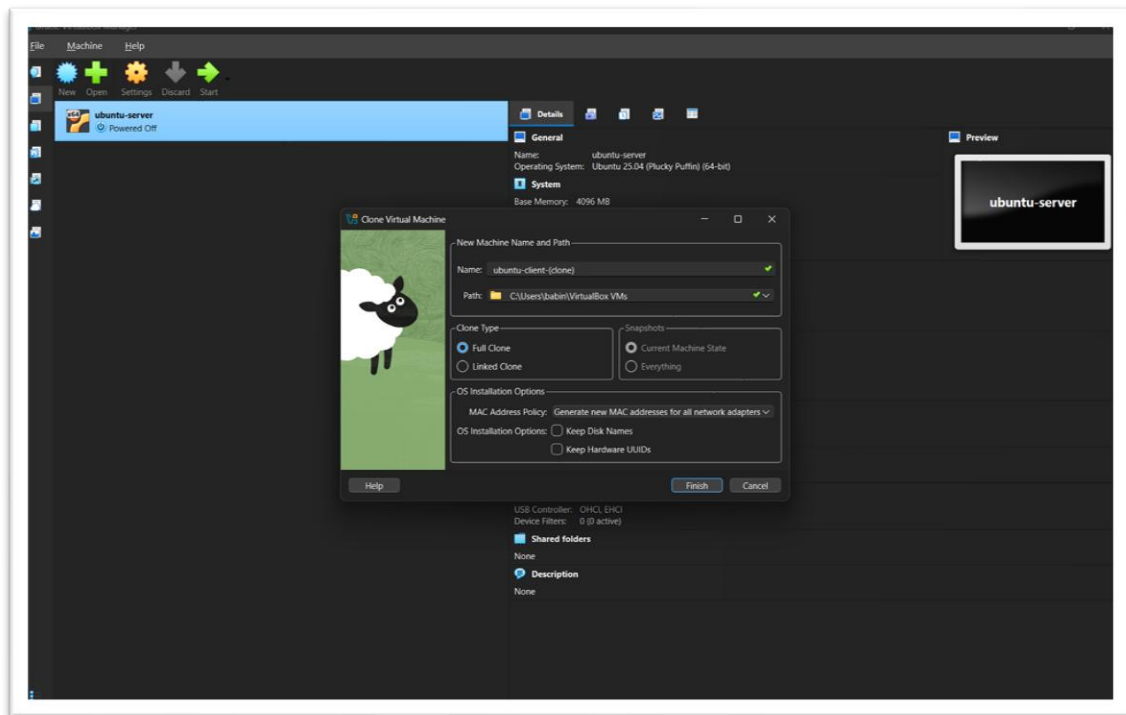


Figure 2 — VirtualBox Clone Virtual Machine dialog showing Full Clone and new MAC address generation.

Step 3 — Create a NAT Network

Open VirtualBox Manager's Network section and create/configure a NAT Network named Dummy. The captured configuration uses the IPv4 network 192.168.50.0/24 with DHCP enabled.

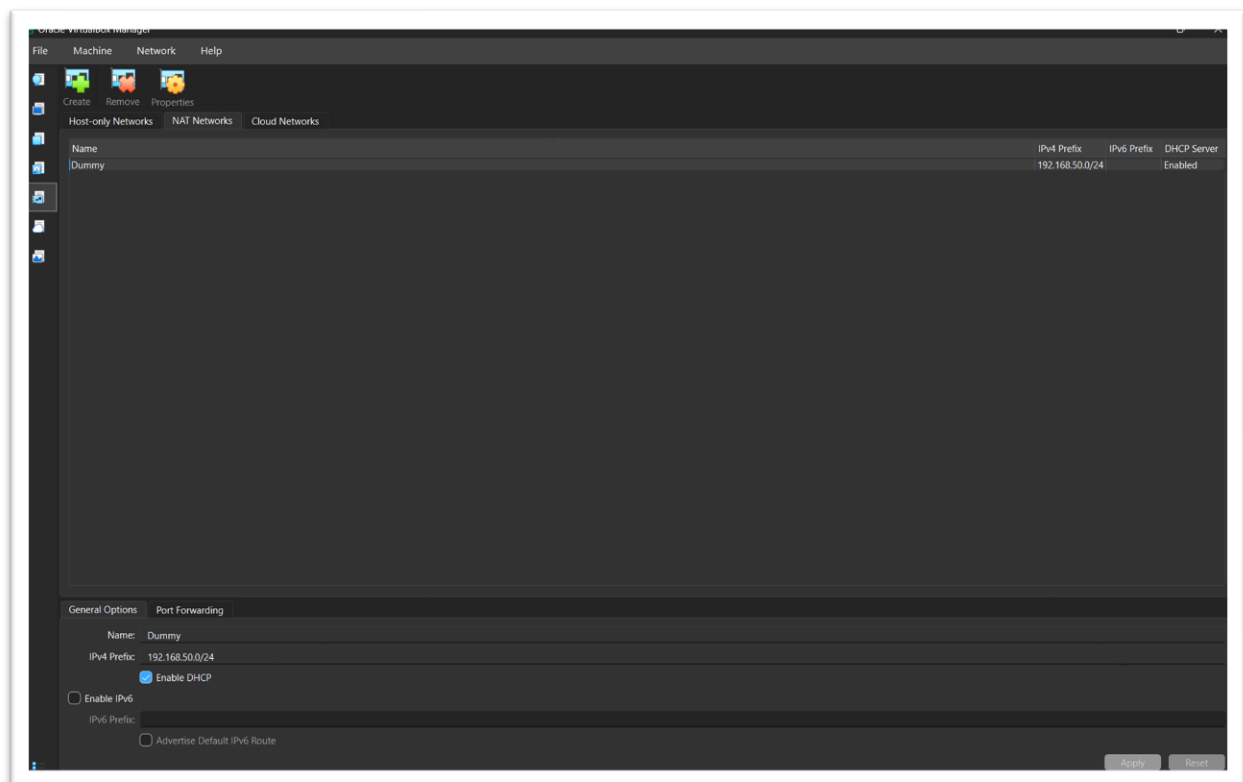


Figure 3 — VirtualBox NAT Network configuration showing Dummy, 192.168.50.0/24 and DHCP enabled.

Step 4 — Configure the Ubuntu Server Network Adapter

Open the Ubuntu server VM settings, select Network → Adapter 1, enable the network adapter, and attach it to the NAT Network named Dummy. Keep the virtual cable connected.

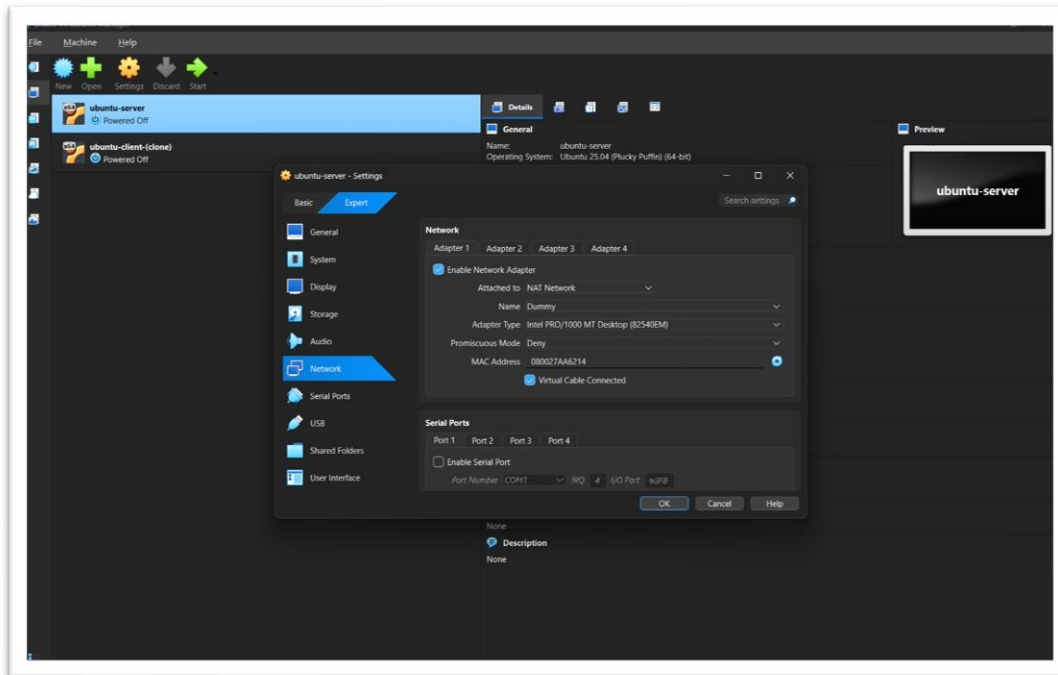


Figure 4 — Ubuntu server network adapter configured for the Dummy NAT Network.

Step 5 — Configure the Ubuntu Client Network Adapter

Open the Ubuntu client VM settings and configure Adapter 1 in the same way. Attach it to the Dummy NAT Network and keep the virtual cable connected. Both virtual machines must use the same NAT Network for direct communication.

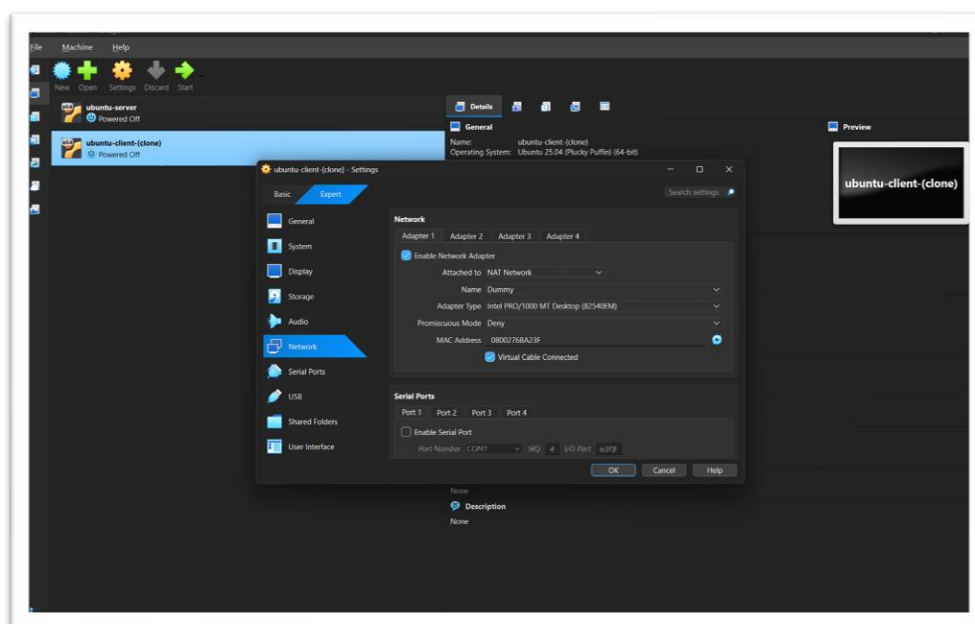


Figure 5 — Ubuntu client network adapter configured for the Dummy NAT Network.

Step 6 — Check the IP Address of Each VM

Start both VMs and run the following command in each Ubuntu terminal to display the network interfaces and assigned IPv4 address:

```
ip a
```

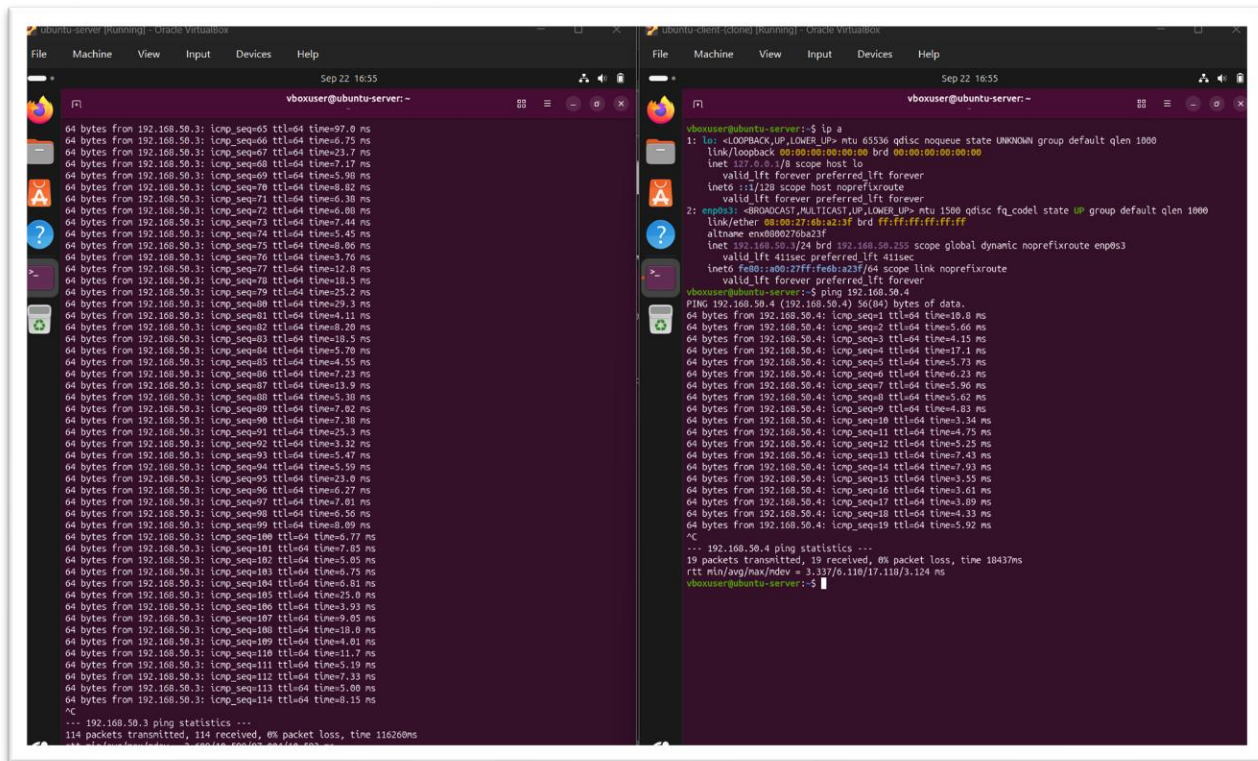
The screenshots show the VMs receiving addresses from the 192.168.50.0/24 network. The two endpoints used for the connectivity test are 192.168.50.3 and 192.168.50.4.

Step 7 — Test Connectivity from the Ubuntu Server

From the Ubuntu server terminal, send ICMP echo requests to the client VM:

```
ping 192.168.50.4
```

The captured output shows replies from 192.168.50.4 and reports 0% packet loss for the completed test.



The image contains two side-by-side terminal screenshots from an Ubuntu server. The left terminal shows the output of the 'ip a' command, displaying network interfaces 'lo' and 'eth0' with their respective IP addresses (127.0.0.1 and 192.168.50.3). The right terminal shows the output of the 'ping 192.168.50.4' command, displaying 19 successful ping requests with 0% packet loss and a round-trip time of approximately 3.137ms.

Figure 6 — Successful ping from the Ubuntu server to 192.168.50.4.

Step 8 — Test Connectivity from the Ubuntu Client

From the other VM, test connectivity back to the server using:

```
ping 192.168.50.3
```

The captured output shows replies from 192.168.50.3, demonstrating two-way IP connectivity between the VMs.

